

MODELING SPOT MICROPHONE SIGNALS USING THE SINUSOIDAL PLUS NOISE APPROACH

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ABSTRACT

This paper focuses on high-fidelity multichannel audio coding based on an enhanced adaptation of the well-known sinusoidal plus noise model (SNM). Sinusoids cannot be used per se for high-quality audio modeling because they do not represent all the audible information of a recording. The noise part has also to be treated to avoid an artificial sounding resynthesis of the audio signal. Generally, the encoding process needs much higher bitrates for the noise part than the sinusoidal one. Our objective is to encode spot microphone signals using the SNM, by taking advantage of the inter-channel similarities to achieve low bitrates. We demonstrate that for a given multichannel audio recording, the noise part for each spot microphone signal (before the mixing stage) can be obtained by using its noise envelope to transform the noise part of just one of the signals (the so-called "reference signal", which is fully encoded).

1. INTRODUCTION

In the last few years, multichannel audio began gradually to displace stereophonic sound systems because it offers significant advantages to audio reproduction when compared to stereo sound¹. The large number of channels gives to the listener the sensation of being "surrounded" by sound and immerses him with a realistic acoustic scene. The main problem with the increased number of channels is the demand on higher datarates for storage and transmission purposes. Low-bandwidth applications (such as Internet streaming and wireless transmission) remain demanding, although coding methods (MPEG AAC, Dolby AC-3, etc.) achieve significant coding gain. This paper focuses on reducing the transmission (and storage) requirements of spot microphone signals *before* those are mixed into the final multichannel audio mix, by exploiting the similarities between such signals of the same multichannel recording. We mention that multichannel audio coding methods that exploit interchannel redundancy have been proposed in the past, including Mid/Side Coding [1] (for frequencies below 2 kHz), Intensity Stereo Coding [2] (for frequencies above 2 kHz), and KLT-based methods [3].

The concept of Spatial Audio Coding (SAC) has been introduced with the objective of further taking advantage of interchannel redundancies and irrelevancies in multichannel audio recordings. In order to achieve low bitrate coding, SAC captures the spatial image of a multichannel audio signal with a compact set

of parameters. The goal is to resynthesize the original multichannel spatial image at the decoder by encoding only one channel of audio (reference channel) and the set of parameters as side information. One of the most popular implementations within SAC is Binaural Cue Coding (BCC) [4]. BCC encodes as additional information the subband interchannel level difference, time difference, and correlation of each channel with respect to the reference audio channel, achieving bitrates in the order of few KBits/sec for the side information of each channel. Parametric Stereo (PS), operates in very similar philosophy [5].

The most common method of producing a multichannel audio recording in its final form is the mixing of a large number of microphone signals that are placed in a venue for recording a music performance. Interactive applications that are of immense interest for immersive audio environments, such as remote mixing of the multichannel recording and remote collaboration of geographically distributed musicians [6], can be accomplished only when the decoder has access to the microphone signals and locally creates the final mix. For these applications, the number of multiple audio channels to be encoded is much higher than in multichannel recordings, and low bitrate encoding of each channel is a very critical aspect. In this paper, we focus on the spot microphone signals of a multichannel recording, which are microphones that are placed close to the sound source. The recordings of these microphones heavily depend on the instruments that are near the microphone and not so much on the hall acoustics; these recordings recreate the sense that the sound source is not a point source but rather distributed such as in an orchestra. Hence, resynthesizing the signals captured by these microphones involves enhancing certain instruments and diminishing others, which in most cases overlap in the time and frequency domains.

In this paper, we introduce and validate a novel SNM plus noise transplantation approach for encoding the multiple microphone signals of a music performance with moderate datarate requirements. This would allow for transmission through low bandwidth channels such as the current Internet infrastructure, and for broadcasting over wireless networks. Our method focuses on the microphone signals of a performance *before they are mixed*, and thus can be applied to applications such as remote mixing and distributed performances. In principle, our method attempts to model each microphone signal with respect to a reference audio channel, so in this sense it follows the SAC philosophy. We employ the sinusoids plus noise model for each microphone signal, and we model the signal with the sinusoidal parameters (harmonic part) and the short-time spectral envelope of the noise (modeling noise part). For resynthesis of each microphone signal, we add the harmonic part that was fully encoded, to the noise part which

* This work has been funded by the Greek General Secretariat for Research and Technology, Program EIIAN Code 05NON-EU-1, and by the Marie Curie TOK "ASPIRE" grant within the 6th European Community Framework Program.

¹In this paper the term stereophonic sound refers to 2-channel stereo.

is recreated by using the corresponding noise envelope with the noise residual obtained from the reference channel. This procedure, which we term as *noise transplantation*, is based on the observation that the noise signals of the various channels of the same multichannel recording are very similar when the harmonic part has been captured with an appropriate number of sinusoids [7]. To our knowledge, this is the first attempt to tailor and apply this model to the specific case of multichannel audio, with the final objective of low bitrate high-fidelity multichannel audio coding.

2. BACKGROUND INFORMATION

2.1. Sinusoids Plus Noise Model

The sinusoidal model represents a harmonic signal $s(n)$ as the sum of a small number of sinusoids with time-varying amplitudes and frequencies

$$s(n) = \sum_{l=1}^L A_l(n) \cos(\theta_l(n)), \quad (1)$$

where $A_l(n)$ and $\theta_l(n)$ is the instantaneous amplitude and phase, respectively. Several variations of the sinusoids plus noise model have been proposed for applications such as signal modification and low bitrate coding, focusing on three different problems: (1) accurately estimating the sinusoidal parameters from the original spectrum (e.g. [8, 9]), (2) representing the modeling error (noise component), and (3) representing signal transients. Here, we focus on the problem of noise representation. In music, a harmonic plus noise model was first proposed in [10]. More recent is the work in [11], where multiresolution analysis was applied for better estimating the sinusoidal parameters. Regarding the noise part, it was not parametrically modeled for best audio quality. The work in [12] and more recently [13] has focused in the noise part modeling. In the first approach, the noise is modeled using a filterbank based on the human auditory system. In the second method, the noise is modeled by applying LPC in the perceptual domain and representing only noise components that are of perceptual relevance. While these noise modeling methods offer the advantage of low bitrate coding for the noise part, the resulting audio quality is usually worse than the quality of the original audio signal (subjective results with average grades around 3.0 in a 5-grade scale have been reported). In our case, we are interested in high-quality audio modeling (achieving a grade around 4.0 is desirable). Our objective is to provide a proof of concept for the noise transplantation procedure and show that indeed this method results in good audio quality compared not only to the sinusoids-only model but also compared with the original recording.

The sound representation is obtained by restricting the sinusoids to modeling only the deterministic part of the sound, leaving the rest of the spectral information in the noise component $e(n)$, i.e., for each short-time frame the signal can be represented as

$$s(n) = \sum_{l=1}^L A_l(n) \cos(\theta_l(n)) + e(n). \quad (2)$$

After the sinusoidal parameters are estimated, the noise component is computed by subtracting the harmonic component from the original signal. In this paper, we model the noise component of the sinusoidal model as the result of filtering a residual noise component with an autoregressive (AR) filter that models the noise spectral envelope. Linear Predictive (LP) analysis is applied to estimate the spectral envelope of the sinusoidal noise. In other

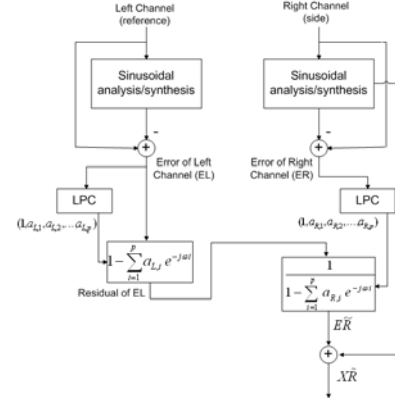


Figure 1: *Noise transplantation.* The LPC residual of the reference signal's noise component is filtered by the side signal's noise envelope and added to its sinusoidal component.

words, we assume the following equation for the noise component of the sinusoidal model

$$e(n) = \sum_{i=1}^p \alpha(i) e(n-i) + r_e(n). \quad (3)$$

The quantity $e(n)$ is the sinusoidal noise component, while $r_e(n)$ is the residual of the noise and p is the AR filter order. The $p+1^{th}$ -dimensional vector $\vec{\alpha}^T = (1, -\alpha_1, -\alpha_2, \dots, -\alpha_p)$ represents the spectral envelope of the noise component $e(n)$. In the frequency domain (3) becomes

$$S_e(e^{j\omega}) = \left| \frac{1}{A(e^{j\omega})} \right|^2 S_{r_e}(e^{j\omega}), \quad (4)$$

where $S_e(e^{j\omega})$ and $S_{r_e}(e^{j\omega})$ is the power spectrum of $e(n)$ and $r_e(n)$, respectively, while $A(e^{j\omega})$ is the frequency response of the LP filter $\vec{\alpha}$. Since in this paper there are two noise quantities introduced, i.e., the sinusoidal model noise e and its whitened version r_e , we will refer to e as the (sinusoidal) noise signal and to r_e as the residual (noise) of e . In the remainder of the paper, the sinusoidal model employed follows the procedure described in [10]. For convenience, we refer to this model as the Sinusoids plus Noise Model (SNM).

3. NOISE TRANSPLANTATION

We start by considering two spot microphone signals of a music performance, in which the two microphones are placed close to two distinct groups of instruments of the orchestra. The first microphone signal is denoted by $x_L(n)$ (for simplicity we refer to this signal as the left channel, which should not be confused with the channels of the multichannel mix), while the second one is denoted by $x_R(n)$ (referred to as the right channel). Each of these microphone signals mainly captures the sound from the closest group of instruments, but also captures the sound from all the other instruments of the orchestra (this is especially true for live concert hall performances). Thus, the two recordings are similar in content, and this is apparent in most multichannel recordings in such settings. Alternatively, one of the channels (the reference signal) could be a sum signal of all the spot recordings.

Sinusoidal models capture the harmonics of the original audio signal well if the number of harmonics used is carefully cho-

sen. However, especially for music signals, the harmonic component is not sufficient for high-quality synthesis; its structured nature and the lack of “randomness” in the signal is audible even if a high number of sinusoids is used. The noise signal e , which contains the spectral information which is considered of random nature, is necessary for high-quality audio synthesis. It mostly contains higher-frequency information, and adds the acoustically needed “randomness” to the sinusoidal component. In coding applications, the noise signal will require a much higher degree in terms of datarates compared to the sinusoidal component, exactly due to its quasi-random nature. Thus, we are interested here to propose a model that is based on the sinusoidal component of the audio signal, but can result in high-quality audio synthesis at the decoder.

In order to achieve this objective, we propose a scheme that is similar to the Spatial Audio Coding philosophy. In other words, we propose that given a collection of microphone signals that correspond to the same multichannel recording (and thus have similar content), we encode as a full audio channel only one of the signals (reference signal). We model the remaining signals with the SNM model, retaining their sinusoidal components and the noise spectral envelope (filter $\tilde{\alpha}$ in (3)). For resynthesis, we model the reference signal with the SNM in order to obtain its noise signal e , and from it we obtain the LP residual r_e using LPC analysis. Finally, we reconstruct each microphone signal using its sinusoidal component and its noise LP filter; its sinusoidal component is added to the noise component that we obtain by filtering with the signal’s LP noise shaping filter the LPC residual of the sinusoidal noise from the reference signal. The assumption is that, as the harmonics capture most of the important information for each microphone signal and the LP coefficients capture most of the channel-specific noise characteristics, the residual noise part that remains will be similar for all the microphone signals. This assumption is in fact verified in Section 4. By taking the reference residual (whitened sinusoidal noise) and filtering it with the correct noise envelope (the envelope of side channel k , where the reference and side signals must be time-aligned), we can obtain a noise signal with very similar spectral properties to the initial noise component of the side channel k . This procedure is depicted in the diagram of Fig. 1.

To formalize the previous discussion, considering a multichannel recording with M microphone signals, we introduce the general relation for the resynthesis of one of the *side* microphone signals x_k (as opposed to the *reference* signal $x_{(ref)}$),

$$\hat{x}_k(n) = \sum_{l=1}^L A_{k,l}(n) \cos(\theta_{k,l}(n)) + \hat{e}_k(n), \quad k = 1, \dots, M, \quad (5)$$

where $\hat{e}_k(n)$ is represented in the frequency domain as

$$\hat{S}_{e_k}(e^{j\omega}) = \left| \frac{1}{1 - \sum_{i=1}^p \alpha_k(i) e^{-j\omega i}} \right|^2 S_{r_{e(ref)}}(e^{j\omega}), \quad (6)$$

In the equations above, $A_{k,l}(t)$ and $\theta_{k,l}(t)$ are the estimated sinusoidal parameters of microphone signal k , $\{\alpha_k\}$ is the signal’s LP noise shaping filter, while $\hat{e}_k(n)$ is the estimated noise component using the noise transplantation procedure described. The residual of the noise component of the reference signal can be found as

$$S_{r_{e(ref)}}(e^{j\omega}) = \left| 1 - \sum_{i=1}^p \alpha_{(ref)}(i) e^{-j\omega i} \right|^2 S_{e_{(ref)}}(e^{j\omega}). \quad (7)$$

Thus, $S_{r_{e(ref)}}(e^{j\omega})$ is the power spectrum of the reference signal noise residual (AR modeling error of the sinusoidal noise), and $e_{(ref)}$ is the sinusoidal noise obtained from the reference.

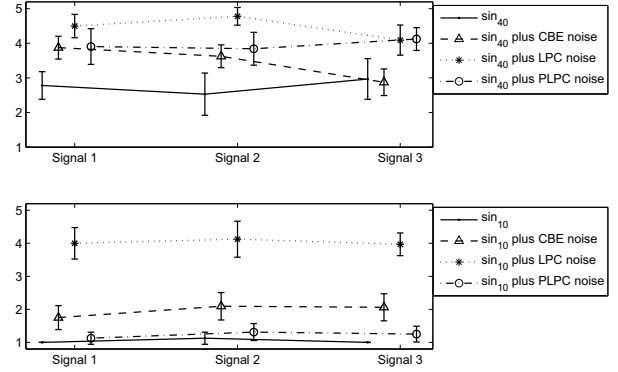


Figure 2: Results from the quality rating DCR listening tests corresponding to sinusoidal modeling with (a) 40 sinusoids per frame (top), and (b) 10 sinusoids per frame (bottom).

4. EXPERIMENTAL RESULTS

In this section, we evaluate the performance of our method by conducting two different experiments. In the first case, we used two microphone signals of a multichannel recording of a concert hall performance, where one of the microphones captures mainly the female voices of the orchestra’s chorus and is used here as the side channel, while the other one mainly captures the male voices and is used as the reference signal. The example examined here is proposed based on the fact that our goal is to resynthesize each microphone signal independently of the others, with the use of only one reference signal and the model parameters (sinusoids and LP filter) that characterize the side microphone signal. The two recordings used here were chosen based on the fact that they have been used in our previous experiments with other modeling methods [14]. In order to compare the quality of the resynthesized (side) signal with respect to the original microphone recording, we conducted three different DCR-based (Degradation Category Rating) listening tests, using a five-grade scale from 1 (very annoying perceived quality) to 5 (not perceived difference in quality) [15]. Sixteen listeners participated in the listening test individually (the authors are not included), under the same environmental conditions, using high-quality headphones. From the two concert hall recordings, we chose three different parts of the performance of about 10 sec duration each (referred to as Signals 1-3 in Fig. 2).

In the second experiment, we are interested to examine the resynthesis quality of the various spot signals from a downmix sum signal. This is important in cases when spot signals do not contain very similar audio content, which is often the case in studio recordings. We used seven audio signals for the test. Each sound file used was a sum of two original recordings, and more specifically the following signals were created: (1) bass² plus soprano singer², (2) guitar³ plus rock singer³, (3) harpsichord² plus violin², (4) female⁴ plus male speech⁴, (5) trumpet² plus violin, (6) violin plus guitar, and (7) violin plus harpsichord. These seven signals correspond 1-1 to Signals 1-7 in the DCR results depicted in Fig. 3. The instrument that is referred first in the above list is the instrument (side signal) that we wanted to resynthesize from

²<http://sound.media.mit.edu/mpeg4/audio/sqam/>

³courtesy of rock band “Orange Moon”

⁴<http://www.cslu.ogi.edu/corpora/voices/>

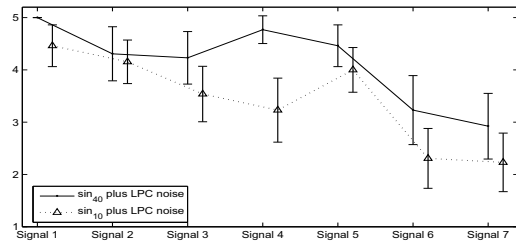


Figure 3: Results from the quality rating DCR listening tests for the downmix case, corresponding to sinusoidal modeling with (a) 40 sinusoids per frame (solid line), and (b) 10 sinusoids per frame (dotted line).

the sum signal (reference signal). 13 volunteers (the authors are not included) participated at this DCR listening test.

The particular implementations in this section are based on a 20 msec analysis/synthesis frame for the sinusoidal model with 50% overlapping (with overlap-add synthesis). The LP order for the AR noise shaping filters is 20. The sampling rate for the recordings used is 44.1 kHz, except for the speech signals⁴ which have 22 kHz rate.

In each plot of the Fig. 2, the solid line corresponds to the sinusoidal model resynthesis (“sin”), the dotted line to our proposed model (“sin plus LPC noise”), the dashed line corresponds to adding the noise (obtained with the Critical Band Energy (CBE) approach [12]) of the side signal to the sinusoidal part of the side signal (“sin plus CBE noise”), while the dashed-dotted line corresponds to adding to the sinusoidal part of the side signal the noise of the reference signal (with PLPC [13] noise shaping model, “sin plus PLPC noise”). A graphical representation of the 95% confidence interval is also given by the two horizontal lines above and below the mean value. From Fig. 2 we can see that the three noise-based methods are superior in comparison to the model based on sinusoidal parameters only, thus, the noise has to be treated to achieve high-quality resynthesis. Both CBE and PLPC approaches give slightly worse results compared to our method, in the case of 40 sinusoids. However, in the 10 sinusoids case, our LPC-based method still achieves a grade around 4.0, while the two other noise-based methods achieve a grade below 2.0. This can be attributed to the fact that PLPC and CBE methods treat the envelope of the noise part only, while our method provides a residual signal for the noise as well (based on similarities among the various spot microphone signals). Finally, it is important to note that, since the resynthesis efficiency of the proposed shaping approach remains almost constant regardless of the number of sinusoids used, we can achieve the final objective of decreasing the bitrate needed for encoding the sinusoidal component.

From the results of the third DCR test shown in Fig. 3, we can notice that in the 40 sinusoids case, the Signals 1-5 achieve a grade above 4.0, while the Signals 6-7 achieve a grade around 3.0 because the percussive sounds cannot be adequately modeled by the SNM, and significant information remains in the residual. Besides, in the 10 sinusoids case, the grade of the Signals 1-2 and Signal 5 is above 4.0.

We, also, conducted an ABX test using the Signals 1-7, of the second experiment, for the case of 40 and 10 sinusoids, respectively. In the ABX test, each listener was presented with the original two instrument recordings that were used to obtain the sum

signal as signals A and B (in random order), as well as the resynthesized signal (Signal X), and was asked to associate X with A or B depending on which instrument prevails in the recording. The total ABX score in both cases was 100%, which means that no crosstalk is introduced using our method.

5. CONCLUSIONS

In this paper, we presented a sinusoids plus noise model that is specifically tailored for multichannel audio, with the objective of low bitrate coding by taking advantage of the similarities among the various spot microphone signals. Our approach offers the possibility of employing the flexible sinusoidal model into low bitrate multichannel audio coding, following a similar SAC philosophy. At the same time, by focusing on the spot signals before those are mixed into the final multichannel mix, our method allows for many applications that are not feasible if the spot signals are not available to the decoder. In the future, we intend to examine the bitrates that are possible to achieve with this model, and to improve the system by better modeling of the transient signals and by using multiresolution analysis.

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